

SPIDER

447 series
218 patients
4 hospitals

expert masks:
vertebrae,
discs, canal

vendor, field
(DICOM)

R1, R2 intensity

signal inside masks:
raw (R1); ratios,
z scores (R2)

R3 mask geometry

volumes, spacing,
length, counts,
voxel volume (mm)

R4 graph-relational

Laplacian spectrum
of spine graph +
scale-free ratios



V vertebra
D disc
C canal

Parallel analyses (same protocol)

Scanner predictability

random forest,
patient grouped CV;
AUC, bootstrap,
permutation

Acquisition controls

voxel size, coverage,
sequence, fixed field,
case mix, UMC,
ComBat

Downstream utility

degeneration and
narrowing: within
domain and across
vendors

Series network

k-NN graph over R3:
assortativity,
communities,
patient-level null